

Assignment-2

1. An HDFS block size is 128 MB. A file of size 640 MB is stored with replication factor 3. How many total blocks are stored in HDFS?
 - a. 5
 - b. 10
 - c. **15**
 - d. 20
2. If HDFS replication factor is 3, and a file occupies 2 GB logical space, what is the actual disk space used?
 - a. 2 GB
 - b. 4 GB
 - c. **6 GB**
 - d. 8 GB
3. An HDFS cluster has 12 DataNodes, each storing 100 blocks. Replication factor is 3. How many unique logical blocks are stored?
 - a. **400**
 - b. 1200
 - c. 300
 - d. 3600
4. If one rack contains 5 DataNodes out of 10 total racks, and data is stored across 2 racks, what is the probability both racks fail if 5 racks fail?
 - a. **0.25**
 - b. 0.5
 - c. 0.1
 - d. 0.75
5. NameNode metadata consumes 150 bytes per block. If HDFS stores 2 million blocks, how much RAM is required?
 - a. 150 MB
 - b. **300 MB**
 - c. 450 MB
 - d. 600 MB
6. File size 1.2GB, block size 128MB, replication 2. Total blocks?
 - a. 10
 - b. 18
 - c. **20**
 - d. 24

7. 640MB file, 128MB block → map tasks?

- a. 4
- b. 5**
- c. 6
- d. 10

8. Reducer takes 8s per key, 200 keys → total time?

- a. 800s
- b. 1200s
- c. 1600s**
- d. 2000s

9. 212 mappers & 4 reducers → shuffle operations?

- a. 12
- b. 16
- c. 48
- d. 4**

10. An HDFS cluster uses a 256 MB block size with replication factor 3. A dataset of 10 TB is stored. Each DataNode has 8 TB usable space. How many minimum DataNodes are required to store the dataset safely?

- a. 4
- b. 6
- c. 8**
- d. 10